FOOD ORDERING MANAGEMENT SYSTEM

DBMS PROJECT REPORT

1. SQL CONCEPTS APPLIED

1 INNER JOIN

2 LEFT JOIN

3 RIGHT JOIN

4 Aggregate Functions

5 GROUP BY, HAVING and Subqueries

6 CASE Expression

2. QUERY IMPLEMENTATION

1. Implement INNER JOIN

- **Display customer details along with their orders..**

```
SQL> SELECT C.CUSTOMER_NAME,O.ORDER_ID,O.TOTAL_AMOUNT FROM CUSTOMER C INNER JOIN ORDERS O ON C.CUSTOMER_ID = O.CUSTOMER_ID;

CUSTOMER_NAME
--------------------------------------------------------------------------------
ORDER_ID   TOTAL_AMOUNT
---------- ------------
SNEHA
106                550

VIJAY KUMAR
107               1250

ANITHA
108                480


CUSTOMER_NAME
--------------------------------------------------------------------------------
ORDER_ID   TOTAL_AMOUNT
---------- ------------
SURESH
109                890
```

- **Display order details with payment information.**

```
SQL> SELECT O.ORDER_ID, O.TOTAL_AMOUNT, O.ORDER_DATE, O.CUSTOMER_ID, P.PAYMENT_DATE, P.PAYMENT_MODE, P.PAYMENT_STATUS FROM PAYMENT P JOIN ORDERS O ON P.ORDE
R_ID = O.ORDER_ID;

ORDER_ID   TOTAL_AMOUNT ORDER_DAT CUSTOMER_ID PAYMENT_D PAYMENT_MODE
---------- ------------ --------- ----------- --------- --------------------
PAYMENT_ST
----------
106                 550 23-SEP-26         101 24-SEP-26 UPI
COMPLETED

107                1250 15-MAY-26         102 15-MAY-26 CASH
PENDING

108                 480 09-MAR-26         103 09-MAR-26 NET BANKING
FAILED


ORDER_ID   TOTAL_AMOUNT ORDER_DAT CUSTOMER_ID PAYMENT_D PAYMENT_MODE
---------- ------------ --------- ----------- --------- --------------------
PAYMENT_ST
----------
109                 890 10-MAY-26         104 10-MAY-26 CARD
PENDING

110                 900 10-OCT-26         105 11-OCT-26 NET BANKING
COMPLETED
```

- **Retrieve products purchased by customers.**

```
SQL> SELECT C.CUSTOMER_ID, C.CUSTOMER_NAME, O.ORDER_ID, P.PRODUCT_ID, P.PRODUCT_NAME, OD.QUANTITY, OD.PRICE, OD.QUANTITY* OD.PRICE AS TOTAL_AMOUNT FROM CUST
OMER C JOIN ORDERS O ON C.CUSTOMER_ID = O.CUSTOMER_ID JOIN ORDER_DETAILS OD ON O.ORDER_ID = OD.ORDER_ID JOIN PRODUCT P ON OD.PRODUCT_ID = P.PRODUCT_ID ORDER
 BY C.CUSTOMER_ID, O.ORDER_DATE;

 CUSTOMER_ID CUSTOMER_NAME       ORDER_ID   PRODUCT_ID PRODUCT_NAME        QUANTITY      PRICE TOTAL_AMOUNT
------------ ------------------ ---------- ---------- ------------------ -------- ---------- ------------
         101 SNEHA              106        P01        IDLI                      2     250.00       500.00
         102 VIJAY KUMAR        107        P02        BUTTER NAAN               3     240.00       720.00
         103 ANITHA             108        P03        CHICKEN FRIED RICE        5     250.00      1250.00
         104 SURESH             109        P05        BROWNIE                   2     300.00       600.00
         105 KAMESH             110        P06        CHICKEN BURGER            4     400.00      1600.00
```

Implement LEFT JOIN

- **Display all customers including customers who have not placed orders.**

```
SQL> SELECT C.CUSTOMER_ID, C.CUSTOMER_NAME, C.EMAIL, O.ORDER_ID, O.ORDER_DATE, O.TOTAL_AMOUNT, O.ORDER_STATUS FROM CUSTOMER C LEFT JOIN ORDERS O ON C.CUSTOM
ER_ID = O.CUSTOMER_ID ORDER BY C.CUSTOMER_ID;

 CUSTOMER_ID CUSTOMER_NAME       EMAIL                                    ORDER_ID   ORDER_DAT TOTAL_AMOUNT ORDER_STATUS
------------ ------------------ ---------------------------------------- ---------- --------- ------------ ------------
```

- **Find customer without any purchases;**

```
SQL> SELECT C.CUSTOMER_ID, C.CUSTOMER_NAME, C.EMAIL, C.PHONE FROM CUSTOMER C
 LEFT JOIN ORDERS O ON C.CUSTOMER_ID = O.CUSTOMER_ID WHERE O.ORDER_ID IS NUL
L ORDER BY C.CUSTOMER_ID;

no rows selected
```

- **Display all product including products with no sales;**

```
SQL> SELECT P.PRODUCT_ID, P.PRODUCT_NAME, P.PRICE FROM PRODUCT P LEFT JOIN
  2  ORDER_DETAILS OD ON P.PRODUCT_ID = OD.PRODUCT_ID WHERE OD.PRODUCT_ID IS NULL ORDER BY P.PRODUCT_ID;

no rows selected
```

```
SQL> SELECT O.ORDER_ID, O.ORDER_DATE, O.TOTAL_AMOUNT, O.ORDER_STATUS, C.CUSTOMER_ID, C.CUSTOMER_NAME, C.EMAIL FROM CUSTOMER C RIGHT JOIN ORDERS O ON C.CUSTO
MER_ID = O.CUSTOMER_ID ORDER BY O.ORDER_DATE;

ORDER_ID   ORDER_DAT TOTAL_AMOUNT ORDER_STATUS        CUSTOMER_ID CUSTOMER_NAME    EMAIL
---------- --------- ------------ ------------------ ----------- ---------------- --------------------------------------------------------------------
------------------------------
108        09-MAR-26       480.00 Preparing                  103 ANITHA  anitha@gmail.com
109        10-MAY-26       890.00 Preparing                  104 SURESH  suresh@gmail.com
107        15-MAY-26      1250.00 Delivered                  102 VIJAY KUMAR      Vijay@gmail.com
106        23-SEP-26       550.00 Pending                    101 SNEHA   sneha@gmail.com
110        10-OCT-26       900.00 Delivered                  105 KAMESH  kamesh@gmail.com
```

- **Find orders where customer details are missing;**

```
SQL> SELECT O.ORDER_ID, O.ORDER_DATE, O.TOTAL_AMOUNT, O.ORDER_STATUS FROM CU
STOMER C RIGHT JOIN ORDERS O ON C.CUSTOMER_ID = O.CUSTOMER_ID WHERE C.CUSTOM
ER_ID IS NULL;

no rows selected
```

.

- **Display all payment record with order details:**

```
SQL> SELECT P.PAYMENT_ID, P.PAYMENT_DATE, P.PAYMENT_MODE, P.PAYMENT_STATUS, P.TRANSACTION_AMOUNT, O.ORDER_ID, O.ORDER_DATE, O.TOTAL_AMOUNT AS ORDER_AMOUNT,
O.ORDER_STATUS FROM ORDERS O RIGHT JOIN PAYMENT P ON O.ORDER_ID = P.ORDER_ID ORDER BY P.PAYMENT_DATE;

PAYMENT PAYMENT_D PAYMENT_MODE        PAYMENT_ST TRANSACTION_AMOUNT ORDER_ID
------- --------- ------------------- ---------- ----------------- ----------
ORDER_DAT ORDER_AMOUNT ORDER_STATUS
--------- ------------ -------------------
PAY103  09-MAR-26 NET BANKING         FAILED                 1000 108
09-MAR-26          480 Preparing

PAY104  10-MAY-26 CARD                PENDING                2000 109
10-MAY-26          890 Preparing

PAY102  15-MAY-26 CASH                PENDING                 900 107
15-MAY-26         1250 Delivered


PAYMENT PAYMENT_D PAYMENT_MODE        PAYMENT_ST TRANSACTION_AMOUNT ORDER_ID
------- --------- ------------------- ---------- ----------------- ----------
ORDER_DAT ORDER_AMOUNT ORDER_STATUS
--------- ------------ -------------------
PAY101  24-SEP-26 UPI                 COMPLETED              3000 106
23-SEP-26          550 Pending

PAY105  11-OCT-26 NET BANKING         COMPLETED              1500 110
10-OCT-26          900 Delivered
```

Retrieve complete order details Generate a complete order report by combining: ● Customer Name ● Product Name ● Quantity ● Order Date ● Total Amount ● Payment Status

```
SQL> SELECT C.CUSTOMER_NAME, P.PRODUCT_NAME, OD.QUANTITY, O.ORDER_DATE, O.TOTAL_AMOUNT, PM.PAYMENT_STATUS FROM CUSTOMER C JOIN ORDERS O ON C.CUSTOMER_ID = O
.CUSTOMER_ID JOIN ORDER_DETAILS OD ON O.ORDER_ID = OD.ORDER_ID JOIN PRODUCT P ON OD.PRODUCT_ID = P.PRODUCT_ID LEFT JOIN PAYMENT PM ON O.ORDER_ID = PM.ORDER_
ID ORDER BY O.ORDER_DATE, C.CUSTOMER_ID;

CUSTOMER_NAME
--------------------------------------------------------------------------------
PRODUCT_NAME
--------------------------------------------------------------------------------
  QUANTITY ORDER_DAT TOTAL_AMOUNT PAYMENT_ST
---------- --------- ------------ ----------
ANITHA
CHICKEN FRIED RICE
         5 09-MAR-26          480 FAILED

SURESH
BROWNIE
         2 10-MAY-26          890 PENDING

CUSTOMER_NAME
--------------------------------------------------------------------------------
PRODUCT_NAME
--------------------------------------------------------------------------------
  QUANTITY ORDER_DAT TOTAL_AMOUNT PAYMENT_ST
---------- --------- ------------ ----------

VIJAY KUMAR
BUTTER NAAN
         3 15-MAY-26         1250 PENDING

SNEHA
IDLI

CUSTOMER_NAME
--------------------------------------------------------------------------------
PRODUCT_NAME
--------------------------------------------------------------------------------
  QUANTITY ORDER_DAT TOTAL_AMOUNT PAYMENT_ST
---------- --------- ------------ ----------
         2 23-SEP-26          550 COMPLETED

KAMESH
CHICKEN BURGER
         4 10-OCT-26          900 COMPLETED
```

Display customer purchase history:

- **All products purchased by a customer:**

```
SQL> SELECT C.CUSTOMER_NAME, O.ORDER_ID, O.ORDER_DATE, P.PRODUCT_NAME, OD.QUANTITY, OD.PRICE FROM CUSTOMER C JOIN ORDERS O ON C.CUSTOMER_ID = O.CUSTOMER_ID
JOIN ORDER_DETAILS OD ON O.ORDER_ID = OD.ORDER_ID JOIN PRODUCT P ON OD.PRODUCT_ID = P.PRODUCT_ID WHERE C.CUSTOMER_ID = '105' ORDER BY O.ORDER_DATE;

CUSTOMER_NAME
--------------------------------------------------------------------------------
ORDER_ID   ORDER_DAT
---------- ---------
PRODUCT_NAME
--------------------------------------------------------------------------------
  QUANTITY      PRICE
---------- ----------
KAMESH
110        10-OCT-26
CHICKEN BURGER
         4        400
```

- **Total amount spent by each customer:**

```
SQL> SELECT C.CUSTOMER_ID, C.CUSTOMER_NAME, ROUND(SUM(O.TOTAL_AMOUNT), 2) AS TOTAL_AMOUNT_SPENT FROM CUSTOMER C JOIN ORDERS O ON C.CUSTOMER_ID = O.CUSTOMER_
ID GROUP BY C.CUSTOMER_ID, C.CUSTOMER_NAME ORDER BY TOTAL_AMOUNT_SPENT DESC;

CUSTOMER_ID
-----------
CUSTOMER_NAME
--------------------------------------------------------------------------------
TOTAL_AMOUNT_SPENT
------------------
       102
VIJAY KUMAR
              1250

       105
KAMESH
               900

CUSTOMER_ID
-----------
CUSTOMER_NAME
--------------------------------------------------------------------------------
TOTAL_AMOUNT_SPENT
------------------

       104
SURESH
               890

       101
SNEHA

CUSTOMER_ID
-----------
CUSTOMER_NAME
--------------------------------------------------------------------------------
TOTAL_AMOUNT_SPENT
------------------
               550

       103
ANITHA
               480
```

- **Number of orders placed by each customer.**

```
SQL> SELECT C.CUSTOMER_ID, C.CUSTOMER_NAME, COUNT(O.ORDER_ID) AS NUMBER_OF_ORDERS FROM CUSTOMER C LEFT JOIN ORDERS O ON C.CUSTOMER_ID = O.CUSTOMER_ID GROUP
BY C.CUSTOMER_ID, C.CUSTOMER_NAME ORDER BY NUMBER_OF_ORDERS DESC;

CUSTOMER_ID
-----------
CUSTOMER_NAME
--------------------------------------------------------------------------------
NUMBER_OF_ORDERS
----------------
       105
KAMESH
              1

       102
VIJAY KUMAR
              1

CUSTOMER_ID
-----------
CUSTOMER_NAME
--------------------------------------------------------------------------------
NUMBER_OF_ORDERS
----------------

       101
SNEHA
              1

       103
ANITHA

CUSTOMER_ID
-----------
CUSTOMER_NAME
--------------------------------------------------------------------------------
NUMBER_OF_ORDERS
----------------
              1

       104
SURESH
              1
```

Latest purchase detail of customers.

```
SQL> SELECT C.CUSTOMER_ID, C.CUSTOMER_NAME, O.ORDER_ID, O.ORDER_DATE AS LATEST_PURCHASE_DATE, O.TOTAL_AMOUNT, O.ORDER_STATUS FROM CUSTOMER C JOIN ORDERS O O
N C.CUSTOMER_ID = O.CUSTOMER_ID WHERE O.ORDER_DATE = (SELECT MAX(O2.ORDER_DATE) FROM ORDERS O2 WHERE O2.CUSTOMER_ID = C.CUSTOMER_ID ) ORDER BY CUSTOMER_ID;

CUSTOMER_ID
-----------
CUSTOMER_NAME
--------------------------------------------------------------------------------
ORDER_ID   LATEST_PU TOTAL_AMOUNT ORDER_STATUS
---------- --------- ------------ --------------------
        101
SNEHA
106        23-SEP-26          550 Pending

        102
VIJAY KUMAR
107        15-MAY-26         1250 Delivered

CUSTOMER_ID
-----------
CUSTOMER_NAME
--------------------------------------------------------------------------------
ORDER_ID   LATEST_PU TOTAL_AMOUNT ORDER_STATUS
---------- --------- ------------ --------------------

        103
ANITHA
108        09-MAR-26          480 Preparing

        104
SURESH

CUSTOMER_ID
-----------
CUSTOMER_NAME
--------------------------------------------------------------------------------
ORDER_ID   LATEST_PU TOTAL_AMOUNT ORDER_STATUS
---------- --------- ------------ --------------------
109        10-MAY-26          890 Preparing

        105
KAMESH
110        10-OCT-26          900 Delivered
```

Generate Multi-Table Business Reports

Report 1: Customer Order Report Display: ● Customer name ● Order ID ● Order date ● Order status

```
SQL> SELECT C.CUSTOMER_NAME, O.ORDER_ID, O.ORDER_DATE, O.ORDER_STATUS FROM CUSTOMER C JOIN ORDERS O ON C.CUSTOMER_ID = O.CUSTOMER_ID ORDER BY O.ORDER_DATE;

CUSTOMER_NAME
--------------------------------------------------------------------------------
ORDER_ID   ORDER_DAT ORDER_STATUS
---------- --------- --------------------
ANITHA
108        09-MAR-26 Preparing

SURESH
109        10-MAY-26 Preparing

VIJAY KUMAR
107        15-MAY-26 Delivered


CUSTOMER_NAME
--------------------------------------------------------------------------------
ORDER_ID   ORDER_DAT ORDER_STATUS
---------- --------- --------------------
SNEHA
106        23-SEP-26 Pending

KAMESH
110        10-OCT-26 Delivered
```

Report 2: Sales Report Display: ● Product name ● Quantity sold ● Total revenue

```
SQL> SELECT P.PRODUCT_NAME, SUM(OD.QUANTITY) AS QUANTITY_SOLD, SUM(OD.QUANTITY * OD.PRICE) AS TOTAL_REVENUE FROM PRODUCT P JOIN ORDER_DETAILS OD ON P.PRODUC
T_ID = OD.PRODUCT_ID GROUP BY P.PRODUCT_ID, P.PRODUCT_NAME ORDER BY TOTAL_REVENUE DESC;

PRODUCT_NAME
--------------------------------------------------------------------------------
QUANTITY_SOLD TOTAL_REVENUE
------------- -------------
CHICKEN BURGER
            4          1600

CHICKEN FRIED RICE
            5          1250

BUTTER NAAN
            3           720


PRODUCT_NAME
--------------------------------------------------------------------------------
QUANTITY_SOLD TOTAL_REVENUE
------------- -------------
BROWNIE
            2           600

IDLI
            2           500
```

Report 3: Payment Analysis Report Display: ● Payment mode ● Number of transactions ● Successful payments

```
SQL> SELECT PAYMENT_MODE, COUNT(*) AS NUMBER_OF_TRANSACTIONS, SUM( CASE WHEN
 PAYMENT_STATUS = 'COMPLETED' THEN 1 ELSE 0 END ) AS SUCCESSFUL_PAYMENTS
FROM PAYMENT GROUP BY PAYMENT_MODE ORDER BY NUMBER_OF_TRANSACTIONS DESC;

PAYMENT_MODE         NUMBER_OF_TRANSACTIONS SUCCESSFUL_PAYMENTS
-------------------- ---------------------- ------------------
NET BANKING                               2                  1
CARD                                      1                  0
UPI                                       1                  1
CASH                                      1                  0
```

Report 4: Customer Purchase Analysis Find: Top purchasing customers

```
SQL> SELECT C.CUSTOMER_ID, C.CUSTOMER_NAME, SUM(O.TOTAL_AMOUNT) AS TOTAL_SPENDING FROM CUSTOMER C JOIN ORDERS O ON C.CUSTOMER_ID = O.CUSTOMER_ID GROUP BY C
CUSTOMER_ID, C.CUSTOMER_NAME ORDER BY TOTAL_SPENDING DESC;

CUSTOMER_ID
-----------
CUSTOMER_NAME
--------------------------------------------------------------------------------
TOTAL_SPENDING
--------------
       102
VIJAY KUMAR
         1250

       105
KAMESH
          900

CUSTOMER_ID
-----------
CUSTOMER_NAME
--------------------------------------------------------------------------------
TOTAL_SPENDING
--------------

       104
SURESH
          890

       101
SNEHA

CUSTOMER_ID
-----------
CUSTOMER_NAME
--------------------------------------------------------------------------------
TOTAL_SPENDING
--------------
          550

       103
ANITHA
          480
```

Customer with maximum orders:

```
SQL> SELECT C.CUSTOMER_ID, C.CUSTOMER_NAME, COUNT(O.ORDER_ID) AS NUMBER_OF_ORDERS FROM CUSTOMER C JOIN ORDERS O ON C.CUSTOMER_ID = O.CUSTOMER_ID GROUP BY C
CUSTOMER_ID, C.CUSTOMER_NAME HAVING COUNT(O.ORDER_ID) = (SELECT MAX(ORDER_COUNT) FROM (SELECT COUNT(*) AS ORDER_COUNT FROM ORDERS GROUP BY CUSTOMER_ID));

CUSTOMER_ID
-----------
CUSTOMER_NAME
--------------------------------------------------------------------------------
NUMBER_OF_ORDERS
----------------
       105
KAMESH
             1

       102
VIJAY KUMAR
             1

CUSTOMER_ID
-----------
CUSTOMER_NAME
--------------------------------------------------------------------------------
NUMBER_OF_ORDERS
----------------

       104
SURESH
             1

       103
ANITHA

CUSTOMER_ID
-----------
CUSTOMER_NAME
--------------------------------------------------------------------------------
NUMBER_OF_ORDERS
----------------
             1

       101
SNEHA
             1
```

- **Customers with highest spending**

```
SQL> SELECT C.CUSTOMER_ID, C.CUSTOMER_NAME, SUM(O.TOTAL_AMOUNT) AS TOTAL_SPE
NDING FROM CUSTOMER C JOIN ORDERS O ON C.CUSTOMER_ID = O.CUSTOMER_ID GROUP B
Y C.CUSTOMER_ID, C.CUSTOMER_NAME HAVING SUM(O.TOTAL_AMOUNT) =
  2  (SELECT MAX(TOTAL_SPENDING) FROM (SELECT SUM(TOTAL_AMOUNT) AS TOTAL_SPE
NDING FROM ORDERS GROUP BY CUSTOMER_ID));

CUSTOMER_ID
-----------
CUSTOMER_NAME
--------------------------------------------------------------------------------
----
TOTAL_SPENDING
--------------
       102
VIJAY KUMAR
         1250
```